FillBlanks

Fill-in-the-Blanks Question Generator using Natural Language Processing

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ABSTRACT

The FillBlanks system is an AI-based application designed to automatically generate fill-in-the-blank questions from a given paragraph or text file. The system uses Natural Language Processing (NLP) techniques to analyze the input text and identify important entities such as names, dates, locations, and organizations. These identified entities are then replaced with blanks to generate meaningful questions. The application is implemented using Python and the spaCy NLP library, with a graphical user interface developed using Tkinter. Users can enter a paragraph manually or upload a text file and specify the number of questions to generate. The system then produces fill-in-the-blank questions along with an answer key. This project demonstrates how NLP can be used to automate question generation for educational purposes.

PROBLEM DESCRIPTION

Creating questions manually from study materials is a time-consuming process for teachers and educators. Preparing exercises such as fill-in-the-blank questions requires careful reading and identification of important keywords in the text. This process becomes difficult when dealing with large amounts of content.

The main problem addressed in this project is the **automatic generation of meaningful questions from textual content**. Traditional systems often require manual effort or produce random questions without understanding the context.

The FillBlanks system solves this problem by using **Natural Language Processing techniques** to analyze the text and identify significant entities such as people, places, organizations, and dates. These entities are automatically removed from the sentence and replaced with blanks to create questions.

The goal of the project is to create a simple and interactive system that can assist in **educational content generation**.

LIBRARIES & TOOLS USED

Python

Python is the primary programming language used to implement the system. It is widely used for artificial intelligence and natural language processing applications due to its simplicity and powerful libraries.

spaCy (NLP Library)

spaCy is a popular Natural Language Processing library used for text analysis. In this project, spaCy is used for:

- Tokenization (splitting text into words)
- Sentence segmentation
- Named Entity Recognition (NER)

NER helps identify entities such as:

- Person names
- Dates
- Locations
- Organizations

These entities are used to create fill-in-the-blank questions.

Tkinter

Tkinter is the GUI library used to create the graphical user interface of the application. It provides components such as:

- Text boxes
- Buttons
- Labels
- Scrollable text areas

This allows the user to interact with the system easily.

Text File Handling

Python's built-in file handling system is used to allow users to upload text files as input.

SYSTEM DESIGN / WORKFLOW

The system follows a simple workflow to generate questions.

Step 1 – Input Text

The user enters a paragraph manually or uploads a text file containing text.

Step 2 – NLP Processing

The text is processed using the spaCy NLP model. The model analyses the paragraph and detects named entities.

Step 3 – Entity Identification

Important entities such as:

- PERSON
- DATE
- LOCATION
- ORGANIZATION

are detected.

Step 4 – Question Generation

Each detected entity is replaced with a blank (_____) to create a fill-in-the-blank question.

Step 5 – Display Results

The generated questions and their corresponding answers are displayed in the output section of the interface.

WORKFLOW DIAGRAM

User Input (Paragraph / File)

↓

NLP Processing

↓

Named Entity Recognition

↓

Replace Entities with Blanks

↓

Generate Questions

↓

Display Questions + Answer Key

IMPLEMENTATION DETAILS

The system is implemented using Python with the spaCy NLP model and Tkinter for the user interface.

Step 1 – Loading the NLP Model

The spaCy language model is loaded using:

```
nlp = spacy.load("en_core_web_sm")
```

This model enables the system to perform named entity recognition.

Step 2 – Processing the Text

The input paragraph is processed using the NLP model:

```
doc = nlp(paragraph)
```

The document object contains information about sentences, tokens, and entities.

Step 3 – Detecting Sentences

The system splits the paragraph into individual sentences:

```
for sent in doc.sents:
```

This helps generate questions from each sentence separately.

Step 4 – Named Entity Detection

Entities are extracted from each sentence:

```
for ent in sent.ents:
```

Examples of detected entities include:

- Guido van Rossum (PERSON)
- 1991 (DATE)
- Google (ORGANIZATION)
- India (LOCATION)

Step 5 – Generating Questions

The system replaces detected entities with blanks:

```
blank_sentence = sentence.replace(ent.text, "_____")
```

Example:

Original sentence:

Python was created by Guido van Rossum in 1991.

Generated question:

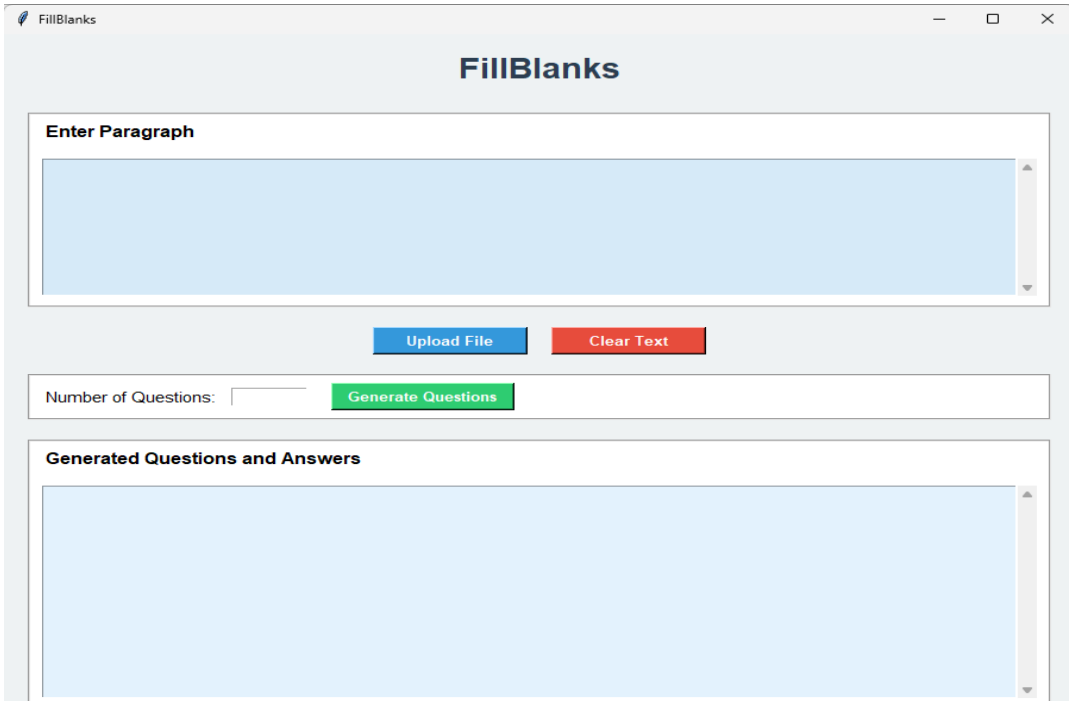
Python was created by _____ in 1991.

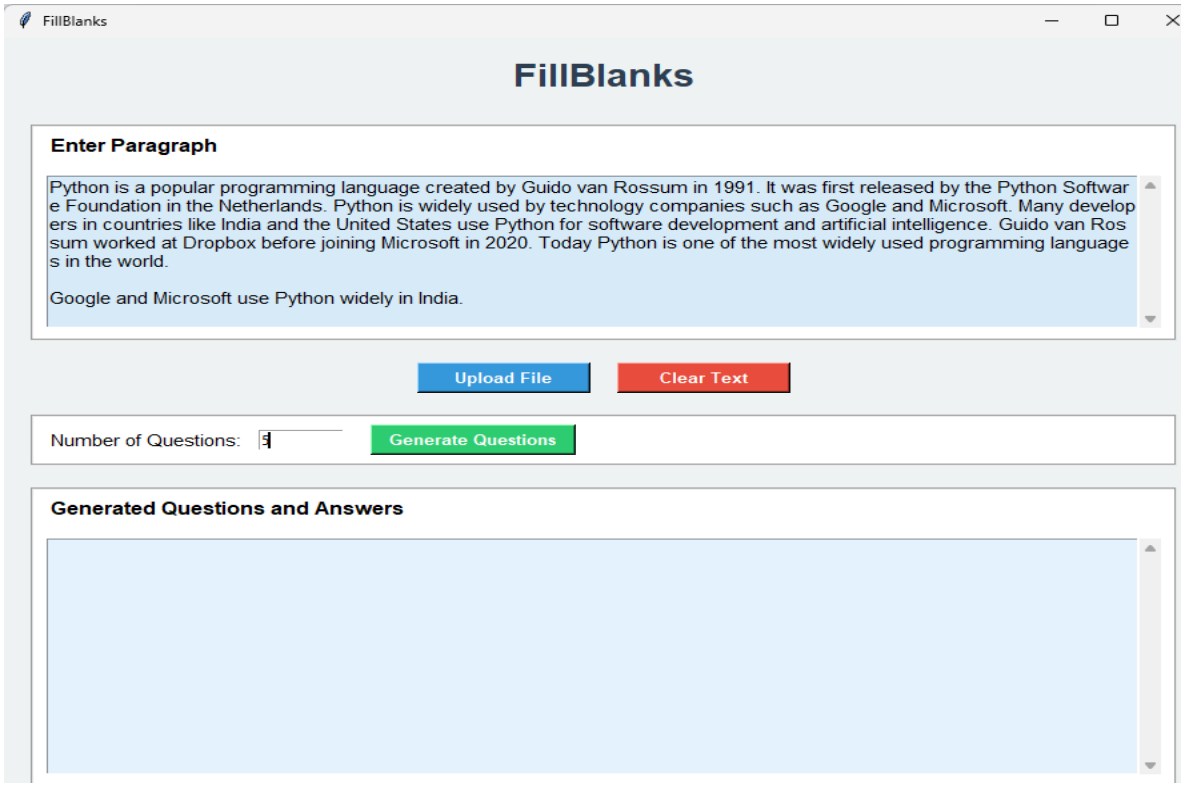
Step 6 – Displaying Output

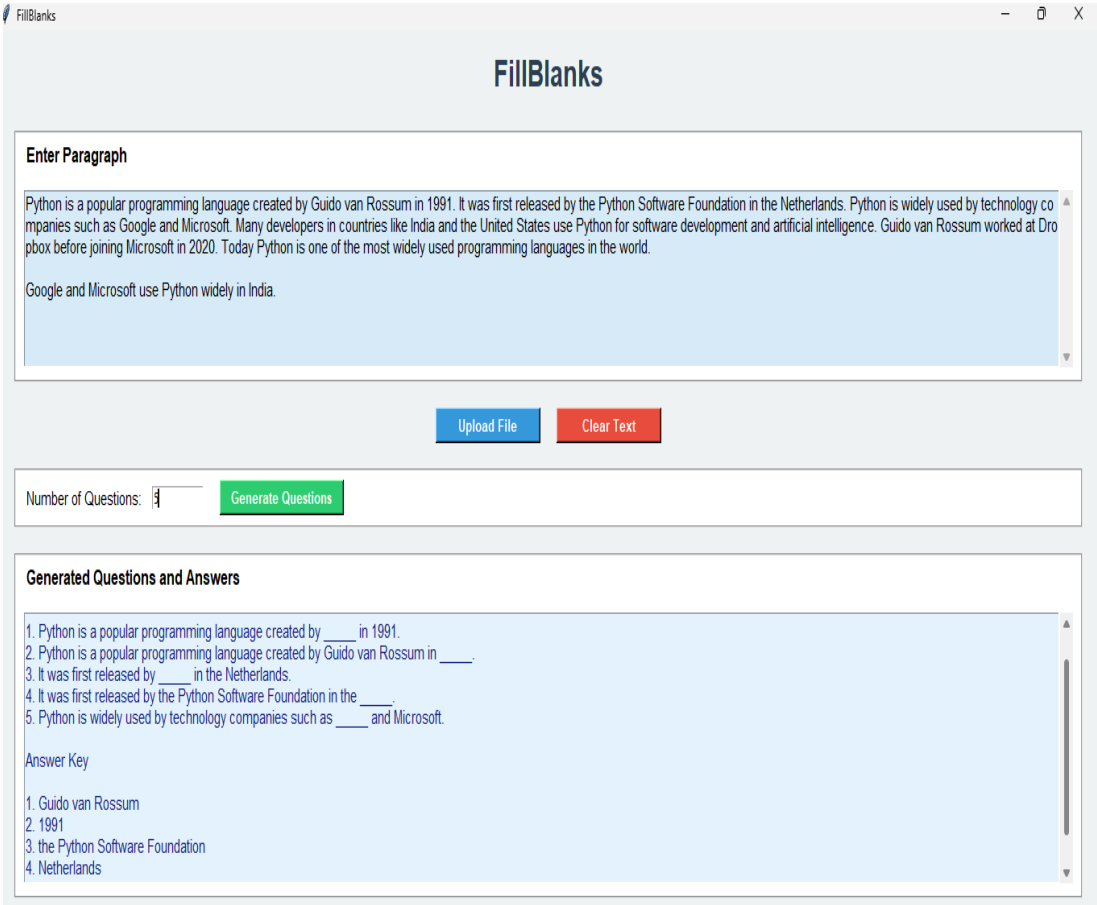
The generated questions and answers are displayed in the output area of the GUI.

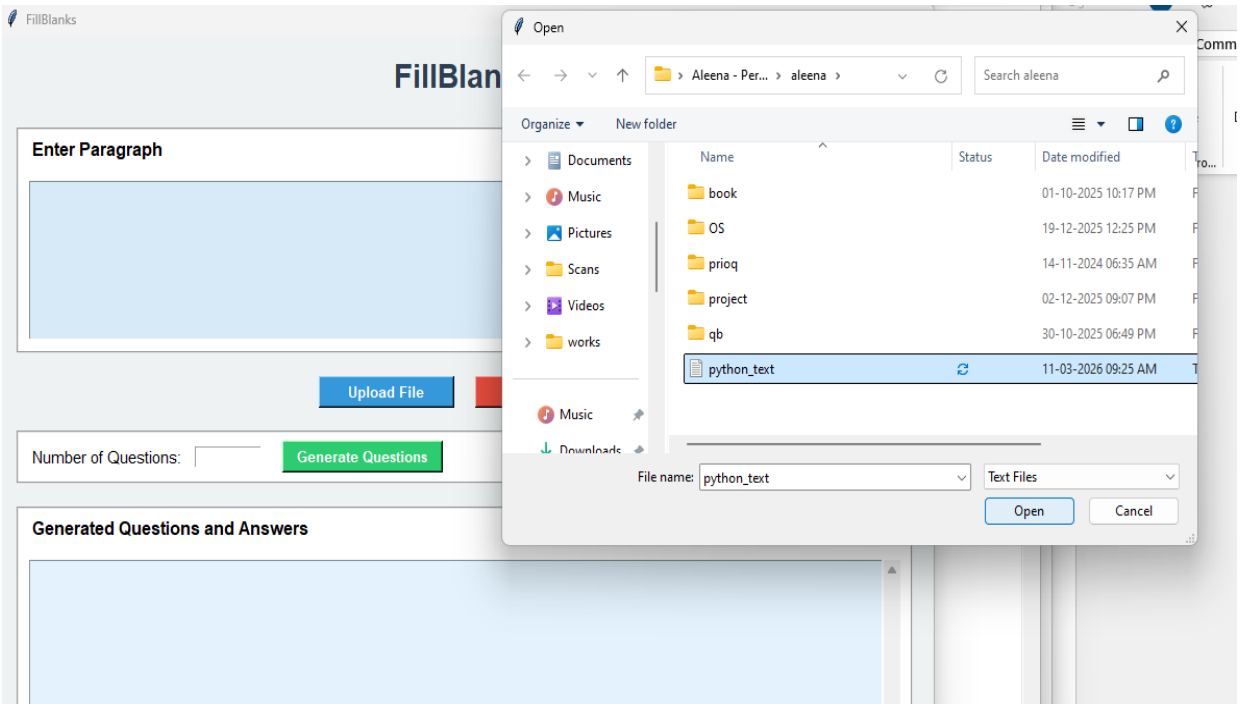
Results & Output

The application successfully generates fill-in-the-blank questions from the input paragraph.






FillBlanks
FillBlanks
Enter Paragraph
Python is a popular programming language created by Guido van Rossum in 1991. It was first released by the Python Software Foundation in the Netherlands. Python is widely used by technology co mpanies such as Google and Microsoft. Many developers in countries like India and the United States use Python for software development and artificial intelligence. Guido van Rossum worked at Dro pbox before joining Microsoft in 2020. Today Python is one of the most widely used programming languages in the world.
Google and Microsoft use Python widely in India.
Upload File
Clear Text
Number of Questions: 5
Generate Questions
Generated Questions and Answers
1. Python is a popular programming language created by ____ in 1991.
2. Python is a popular programming language created by Guido van Rossum in ____.
3. It was first released by ____ in the Netherlands.
4. It was first released by the Python Software Foundation in the ____.
5. Python is widely used by technology companies such as ____ and Microsoft.
Answer Key
1. Guido van Rossum
2. 1991
3. the Python Software Foundation
4. Netherlands


FillBlanks
FillBlan
Enter Paragraph
Upload File
Number of Questions:
Generate Questions
Generated Questions and Answers
Open
Aleena - Per... › aleena ›
Search aleena
Organize
New folder
Documents
Music
Pictures
Scans
Videos
works
Music
Downloads
Name
Status
Date modified
book
01-10-2025 10:17 PM
OS
19-12-2025 12:25 PM
prioq
14-11-2024 06:35 AM
project
02-12-2025 09:07 PM
qb
30-10-2025 06:49 PM
python_text
11-03-2026 09:25 AM
File name: python_text
Text Files
Open
Cancel

Conclusion & Scope for Mini Project Extension

The FillBlanks system successfully demonstrates how Natural Language Processing can be used to automatically generate educational questions from textual content. By using spaCy's named entity recognition capability, the system identifies important entities in the paragraph and replaces them with blanks to create meaningful questions.

The graphical user interface allows users to easily input text, upload files, and generate questions automatically. This project shows how artificial intelligence techniques can simplify the process of question generation.

Future EnhancementsThe system can be further improved by adding additional features such as:

- Automatic generation of **WH questions**
- Support for **multiple question types**
- Integration with **educational platforms**
- Improved NLP models for **better sentence understanding**
- Support for **PDF and document inputs**

These improvements can make the system more powerful and useful in educational environments.